

UCCC 2063 Algorithm Analysis

Assignment
Full marks = 70

Question One: [Asymptotic notations]

Prove $\forall c \exists n_0 \bullet \forall n > n_0 \bullet n^{10000} < cn!$. (10 marks)

[Hint: by default, c is positive real number, and n_0 and n are positive integers.]

Question Two: [Solving Recurrences]

Express your answer in Big-Oh notation. Your answer must be tight.

a) Solve $T(n) = 3T(n/6) + n/\log n$, by master method. (10 marks)

[Hint: Think $T(n) = 3T(n/6) + n$.]

b) Solve $T(n) = 9T(n/6) + n \log^2 n$, by master method. (10 marks)

[Hint: Think $T(n) = 9T(n/6) + n$.]

c) Solve $T(n) = 9T(n/6) + n \log^2 n$, by MI (or substitution). (15 marks)

[Hint: Get the idea from solving $T(n) = 9T(n/6) + n$.]

Question Three: [Taken from Q3, 2023] (25 marks)

(a) We consider rooted binary tree. A leaf node is a node which does not have any children nodes. The height of a binary tree is defined as follows:

The height of an empty tree is defined as 0. For a non-empty tree, the height is defined as the number of nodes in the path from the root to its farthest leaf node.

(i) Give a recursive definition for the height of a binary tree. (5 marks)

(ii) Based on (i), give a recursive algorithm, in pseudocode format, for accepting an input binary tree, and returning its height. (4 marks)

(iii) Analyze the time complexity of your algorithm in (ii). (4 marks)

(b) A programmer uses the following **struct** for constructing a rooted binary tree.

```
struct node {  
    int x;  
    node* left;  
    node* right;  
};
```

Inside a node, **x** is the stored data, **left** and **right** are pointers pointing to the node's left and right children, respectively, if any. If a pointer is not pointing to any node, it is called null pointer (because, the programmer should have assigned it to **NULL**).

Suppose that an n -node binary tree is constructed using the above **struct**. Using induction on n , prove the following statement, for all integers $n \geq 1$.

Statement: An n -node binary tree has $n+1$ null pointers.

(10 marks)

[Hint: The proof should contain base case, and inductive step.]

- (c) A student extends the struct in (b) into the following **struct** for constructing a rooted N -ary tree.

```
struct node {  
    int x;  
    node* child[N];  
};
```

As the **struct** in (b), variable x is the stored data, the pointers are pointing to children nodes, respectively, if any. If a pointer is not pointing to any node, it is a null pointer.

What is the number of null pointers in an n -node N -ary tree, in terms of n and N ?

(2 marks)

[Hint: no need to explain.]

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